TRIANGULATION IN O-MINIMAL STRUCTURES

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1. Introduction

Throughout this lecture, let $\mathcal{S}$ be an o-minimal structure expanding a dense linear order, and all sets and maps are assumed to be $\mathcal{S}$-definable unless stated otherwise. We work in Euclidean space $\mathbb{R}^n$ with its usual topology. We will introduce some terminology in Section 2, and then state and prove the triangulation theorem in Section 3.

2. Basic definitions

2.1. Simplices and complexe.

Definition 2.1 (Simplices)**.** Points $a_0, \dots, a_r \in \mathbb{R}^m$ are *affinely independent* if the vectors $a_1 - a_0, \dots, a_r - a_0$ are linearly independent. In that case the *r-simplex* spanned by $(a_0, \dots, a_r)$ is

$$[a_0, \dots, a_r] := \Big\{ \sum_{i=0}^{r} t_i a_i \;\Big|\; t_i \geq 0,\ \sum_{i=0}^{r} t_i = 1 \Big\} \subset \mathbb{R}^m.$$

Its *vertices* are $a_0, \dots, a_r$. A *face* of $[a_0, \dots, a_r]$ is a simplex $[a_{i_0}, \dots, a_{i_k}]$ obtained by restricting to a subsequence of vertices. A *proper face* is a face different from the simplex itself.

Definition 2.2 (Complexes)**.** A *complex* K in $\mathbb{R}^m$ is a finite collection of simplices in $\mathbb{R}^m$ such that:

(1) if $\sigma \in K$ and τ is a face of σ, then $\tau \in K$;
(2) if $\sigma, \tau \in K$, then $\sigma \cap \tau$ is either empty or a common face of both.

The *polyhedron* (or *geometric realization*) of K is

$$|K| := \bigcup_{\sigma \in K} \sigma \subset \mathbb{R}^m.$$

A *subcomplex* L of K is a subset $L \subset K$ which is a simplicial complex. A complex K is *closed* if it contains all faces of its simplices, i.e. if it is a simplicial complex in the sense above (we keep the word "closed" since it is used in many notes).

Definition 2.3 (Barycenter and barycentric subdivision)**.** Let $\sigma = [a_0, \dots, a_r]$ be a simplex. Its *barycenter* is

$$b(\sigma) := \frac{1}{r+1} \sum_{i=0}^{r} a_i \in \sigma.$$

Let K be a complex. The *barycentric subdivision* K' is the complex whose vertices are the barycenters $b(\sigma)$ of simplices $\sigma \in K$, and whose simplices correspond to strictly increasing chains

$$\sigma_0 \subsetneqq \sigma_1 \subsetneqq \cdots \subsetneqq \sigma_r \quad (\sigma_i \in K)$$

via

$$[b(\sigma_0),\, b(\sigma_1),\, \ldots,\, b(\sigma_r)].$$

Definition 2.4 (Simplicial maps)**.** Let K be a complex in $\mathbb{R}^m$ and L a complex in $\mathbb{R}^n$. A continuous map $\phi : |K| \to |L|$ is *simplicial* if it sends vertices of K to vertices of L and for every simplex $\sigma = [v_0, \ldots, v_r] \in K$ the restriction $\phi|\sigma$ is affine and $\phi(\sigma) = [\phi(v_0), \ldots, \phi(v_r)] \in L$.

2.2. **Triangulations and compatibility.**

Definition 2.5 (Triangulation)**.** Let $A \subset \mathbb{R}^n$ be definable. A *triangulation* of A is a pair (K, h) where K is a finite simplicial complex in some $\mathbb{R}^m$ and $h : |K| \to A$ is a definable homeomorphism.

Definition 2.6 (Compatibility)**.** Let (K, h) be a triangulation of $A \subset \mathbb{R}^n$, and let $A_1, \ldots, A_s \subset A$ be definable subsets. We say (K, h) is *compatible* with $(A_1, \ldots, A_s)$ if for each i the preimage $h^{-1}(A_i)$ is a union of simplices of K (equivalently, $h^{-1}(A_i) = |K_i|$ for some subcomplex $K_i \subset K$).

Remark. Compatibility implies that each A_i is triangulated by the subcomplex K_i and the restriction $h|_{|K_i|}$.

3. The triangulation theorem

Our main goal is to prove the following theorem:

Theorem 3.1. *Let $A \subset \mathbb{R}^n$ be definable and let $A_1, \ldots, A_s \subset A$ be definable subsets. Then there exists a triangulation (K, h) of A which is compatible with $A_1, \ldots, A_s$.*

Before our proof, we need some preparations: firstly we give two extension lemmas, enabling us to extend continuous definable maps from a subcomplex to the whole complex; then we define the multivalued functions, which is a convenient language for doing "piecewise" constructions over simplices.

3.1. **Extension lemmas.**

Lemma 3.2. *Let $\sigma = [a_0, \ldots, a_r] \subset \mathbb{R}^m$ be a simplex and let $\partial\sigma$ be its boundary (the union of proper faces). If $f : \partial\sigma \to \mathbb{R}^k$ is continuous and definable, then there exists a continuous definable extension $F : \sigma \to \mathbb{R}^k$ with $F|\partial\sigma = f$.*

Proof. Let $b = b(\sigma)$ be the barycenter. Every point $x \in \sigma \setminus \{b\}$ lies on a unique segment from b to a boundary point: more precisely, consider the ray

$$\ell_x := \{\, b + t(x - b) : t \geq 0 \,\}.$$

Since σ is compact and convex, ℓ_x intersects $\partial\sigma$ in exactly one point; denote it by $p(x)$. Also, there is a unique $t(x) \in (0, 1]$ such that

$$x = b + t(x)\,(p(x) - b).$$

Define $p(b)$ arbitrarily (say, $p(b) = a_0$) and set $t(b) = 0$.

We first check that $p : \sigma \to \partial\sigma$ and $t : \sigma \to [0, 1]$ are definable and continuous. Definability follows because σ and $\partial\sigma$ are semilinear sets and the above construction can be expressed by quantifying over t and using the linear constraints defining σ. For continuity, note that on each region where the ray from b hits a fixed facet of σ, the maps p and t are given by explicit

rational affine formulas (solving one linear equation), hence continuous; these regions form a finite partition of σ.

Now define $F : \sigma \to \mathbb{R}^k$ by

$$F(x) := \begin{cases} f\big(p(x)\big), & x \neq b, \\ \lim_{x \to b} f(p(x)), & x = b. \end{cases}$$

We claim the limit exists and equals f evaluated at any boundary point? In fact, we instead use a standard cone-extension which avoids any limit: set

$$F\big(b + t(y - b)\big) := f(y) \qquad (y \in \partial\sigma,\ t \in (0, 1]).$$

This is well-defined by uniqueness of the representation $x = b + t(y - b)$ with $y \in \partial\sigma$ and $t \in (0, 1]$. Put $F(b) := f(y_0)$ for some fixed $y_0 \in \partial\sigma$.

To verify continuity at b, note that for $x \to b$ we have $t(x) \to 0$ but $F(x) = f(p(x))$ depends only on the direction. So the above definition is not necessarily continuous unless f is constant. We therefore use the *linear homotopy to a fixed value*. Fix $y_0 \in \partial\sigma$ and define

$$F(x) := \begin{cases} (1 - t(x))\, f(y_0) + t(x)\, f(p(x)), & x \neq b, \\ f(y_0), & x = b. \end{cases}$$

This is definable and continuous: away from b it is a composition of continuous definable maps; and as $x \to b$, we have $t(x) \to 0$, hence $F(x) \to f(y_0) = F(b)$. Finally, if $x \in \partial\sigma$, then necessarily $t(x) = 1$ and $p(x) = x$, so $F(x) = f(x)$. □

Lemma 3.3. *Let K be a (finite) complex and let L be a closed subcomplex of K. If $f : |L| \to \mathbb{R}^k$ is continuous and definable, then there exists a continuous definable extension $F : |K| \to \mathbb{R}^k$.*

Proof. We extend f simplex-by-simplex by induction on the number of simplices of $K \setminus L$. If $K = L$ there is nothing to do.

Choose a simplex $\sigma \in K \setminus L$ of minimal dimension among simplices not in L. Since L is a subcomplex, every proper face of σ lies in L (by minimality). Let $L_1 := L \cup \{\sigma\} \cup \{\text{faces of } \sigma\}$; then L_1 is again a closed subcomplex of K and $|L_1| = |L| \cup \sigma$.

By induction hypothesis it suffices to extend f from $|L|$ to $|L_1|$. On $\partial\sigma \subset |L|$ we already have a continuous definable map $f|\partial\sigma : \partial\sigma \to \mathbb{R}^k$. By Lemma 3.2 there is a continuous definable extension $g : \sigma \to \mathbb{R}^k$ with $g|\partial\sigma = f|\partial\sigma$. Define $f_1 : |L_1| \to \mathbb{R}^k$ by setting $f_1 = f$ on $|L|$ and $f_1 = g$ on σ. These agree on $\partial\sigma$, hence f_1 is well-defined and continuous. Repeat until all simplices of K are added. □

3.2. **Multivalued functions.**

Definition 3.4. Let (A, P) be a triangulated set. A *multivalued function* F on (A, P) is a finite collection of continuous definable functions

$$F = \{\, f_{C,i} : C \in P,\ 1 \leq i \leq k(C) \,\},$$

where $k(C) \geq 0$ and

$$f_{C,1} < \cdots < f_{C,k(C)} \quad \text{on } C.$$

We set

$$\Gamma(F) := \bigcup_{f \in F} \Gamma(f), \qquad F|_C := \{f_{C,i} : 1 \leq i \leq k(C)\},$$

and

$$P^F := \{\Gamma(f) : f \in F\} \ \cup \ \{(f_{C,i}, f_{C,i+1}) : C \in P,\ 1 \le i < k(C)\}.$$

Let

$$A^F := \bigcup_{E \in P^F} E.$$

Then P^F is a partition of $A^F \subset \mathbb{R}^{m+1}$ and

$$\pi(\Gamma(F)) = \pi(A^F) \subset A,$$

where $\pi : \mathbb{R}^{m+1} \to \mathbb{R}^m$ is the projection onto the first m coordinates.

We consider two properties of a multivalued function F on (A, P).

(1) We call F *closed* if for each pair $C, D \in P$ with D a proper face of C and each $f \in F|_C$ there exists $g \in F|_D$ such that

$$g(y) = \lim_{x \to y} f(x) \qquad \text{for all } y \in D.$$

Note that then each $f \in F$ extends continuously to a definable function

$$\mathrm{cl}(f) : \mathrm{cl}(\mathrm{dom}\, f) \to \mathbb{R},$$

the restrictions of $\mathrm{cl}(f)$ to the faces of C belong to F, and $\Gamma(F)$ is closed in $A \times \mathbb{R}$.

(2) We call F *full* if F is closed, $k(C) \ge 1$ for all $C \in P$, and for each pair $C, D \in P$ with D a proper face of C and each $g \in F|_D$ there exists $f \in F|_C$ such that

$$g = \mathrm{cl}(f)|_D.$$

Note that if F is full, then $\pi(\Gamma(F)) = A$, and if A is closed and bounded, then so is A^F.

Lemma 3.5. *Let F be a multivalued function on (A, P). Then F is closed if and only if $\Gamma(F)$ is closed in $A \times \mathbb{R}$.*

Proof. Let $C, D \in P$ with D a proper face of C, and let $f \in F|_C$. Fix $y \in D$ and consider $\lim_{x \to y} f(x)$.

Set

$$l := \liminf_{x \to y} f(x), \qquad L := \limsup_{x \to y} f(x).$$

Clearly $-M \le l \le L \le M$ for some M, and $l = L$ if and only if the limit exists.

Suppose $l < L$. Take any t with $l < t < L$. By definable choice, there are arbitrarily small definable neighborhoods U of y in A such that f takes values both $< t$ and $> t$ on $U \cap C$. Hence $t \in \Gamma(F)$ for all $t \in (l, L)$. This implies $(y, t) \in \Gamma(F)$ for all $t \in (l, L)$. If $\Gamma(F)$ is closed, this is impossible. Hence $l = L$, so $\lim_{x \to y} f(x)$ exists.

Let $g(y)$ be this limit. One checks easily that $\mathrm{cl}(f) : D \to \mathbb{R}$ is continuous. Since $\Gamma(F)$ is closed, $\mathrm{cl}(f) \in F|_D$. This proves that F is closed.

The converse was already noted above. □

Lemma 3.6. *Let (Φ, K) be a triangulation in $\mathbb{R}^m$ of a closed and bounded definable set $A \subset \mathbb{R}^m$, and let F be a full multivalued function on the triangulated set $(A, \Phi^{-1}(K))$. Then (Φ, K) or its barycentric subdivision can be lifted to a triangulation (Ψ, L) in $\mathbb{R}^{m+1}$ of A^F that is compatible with the sets in $\Phi^{-1}(K)^F$.*

Proof. Replacing K by its barycentric subdivision and F by the corresponding set of restrictions, we may assume that the following holds for each $C \in \Phi^{-1}(K)$:

$$(*) \quad \text{if } f_1, f_2 \in F|_C,\ f_1 \neq f_2, \text{ then there exists a vertex of } C \text{ where } \mathrm{cl}(f_1) \neq \mathrm{cl}(f_2).$$

Fix a linear order on $\mathrm{Vert}(K)$. Let $C \in \Phi^{-1}(K)$ with vertices $\Phi^{-1}(a_0), \ldots, \Phi^{-1}(a_k)$, listed in the induced order.

Let $f < g$ be two successive elements of $F|_C$. Put

$$r_i := \mathrm{cl}(f)(\Phi^{-1}(a_i)), \qquad s_i := \mathrm{cl}(g)(\Phi^{-1}(a_i)),$$

and define

$$b_i := (a_i, r_i), \qquad c_i := (a_i, s_i) \in \mathbb{R}^{m+1}.$$

By $(*)$, the hypotheses of Lemma (1.10) are satisfied. Let $L(f, g)$ be the complex constructed in that lemma, and let $\Psi_{f,g} : C \times [0, 1] \to L(f, g)$ be the map defined there. It is a definable homeomorphism.

We also define for each $f \in F|_C$ the closed complex $L(f)$ in $\mathbb{R}^{m+1}$ whose vertices are the points $(a_i, \mathrm{cl}(f)(\Phi^{-1}(a_i)))$.

Now let L be the union of all complexes $L(f, g)$ and $L(f)$, for all $C \in \Phi^{-1}(K)$. A routine verification shows that L is a closed complex in $\mathbb{R}^{m+1}$. The maps $\Psi_{f,g}$ and the maps $\Psi_f(x, \mathrm{cl}(f)(x)) = \Phi(x)$ glue together to a continuous definable map

$$\Psi : A^F \to |L|.$$

One checks that Ψ is bijective, hence a definable homeomorphism.

Using Lemma (1.11), it follows that (Ψ, L) is a triangulation lifting (Φ, K). Compatibility with $\Phi^{-1}(K)^F$ is clear from the construction. □

3.3. **Proof of the triangulation theorem.**

Proof of Theorem 3.1. We proceed by induction on m. The case $m = 0$ is trivial.

Assume the theorem holds in dimension $m - 1$. We first reduce to the case where A is closed and bounded. Let

$$\mu : \mathbb{R} \to (-1, 1), \qquad \mu(t) = \frac{t}{\sqrt{1 + t^2}},$$

and apply μ coordinatewise. Since μ is a definable homeomorphism, it suffices to prove the theorem for the image of A. Thus we may assume from the start that A is closed and bounded.

Let

$$T := \partial A \ \cup\ \partial A_1 \ \cup\ \cdots \ \cup\ \partial A_s.$$

Then T is closed and bounded and satisfies $\dim(T) < m$. By the induction hypothesis, T admits a triangulation compatible with all the sets $T \cap A_i$.

By the good directions lemma, there exists a linear automorphism of $\mathbb{R}^m$ such that the unit vector e_m is a good direction for T. Replacing A and the A_i by their images under this automorphism, we may assume that e_m is a good direction for T.

Set

$$B := \pi(A) = \pi(T) \subset \mathbb{R}^{m-1},$$

where $\pi : \mathbb{R}^m \to \mathbb{R}^{m-1}$ is the projection onto the first $m-1$ coordinates. By cell decomposition, the set T is the disjoint union of graphs of finitely many continuous definable functions on cells of a partition of B. Thus there exists a multivalued function F on (B, P) such that

$$T = \Gamma(F).$$

By the induction hypothesis, B has a triangulation (Φ, K) compatible with all the sets arising from the partition P. Restricting the functions in F to simplices of $\Phi^{-1}(K)$, we obtain a multivalued function on the triangulated set $(B, \Phi^{-1}(K))$. Since T is closed, Lemma 3.5 implies that F is closed. By extending the functions if necessary, we may assume that F is full.

Applying Lemma 3.6, we obtain a triangulation of B^F that is compatible with the sets in $\Phi^{-1}(K)^F$. Finally, since A lies between successive graphs in F, this triangulation induces a triangulation of A which is compatible with $A_1, \ldots, A_s$. □